

PDGen: A Topology Model for (Generative) Shape Models

Enhancing Implicit Shape Generators Using Topological Regularizations

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accelerated binary package for training



Existing Problems

Implicit shape generators with Signed-Distance Functions (SDFs) suffer from an array of problems, mostly of a topological nature:

- **Topological defects** such as broken legs and undesirable thin structures
- **Poor interpolation** capabilities
- **Lack of** topological generalizations or **topological proximity guarantees**



Key Insights | What We Need

- A Probabilistic Model on shape topologies, hence a **topology model as a prior**
- **Synchronize** underlying shape SDFs with topological targets in a differentiable way
- **Convergence guarantee** provided by bounds between the topological distance and the SDF distance

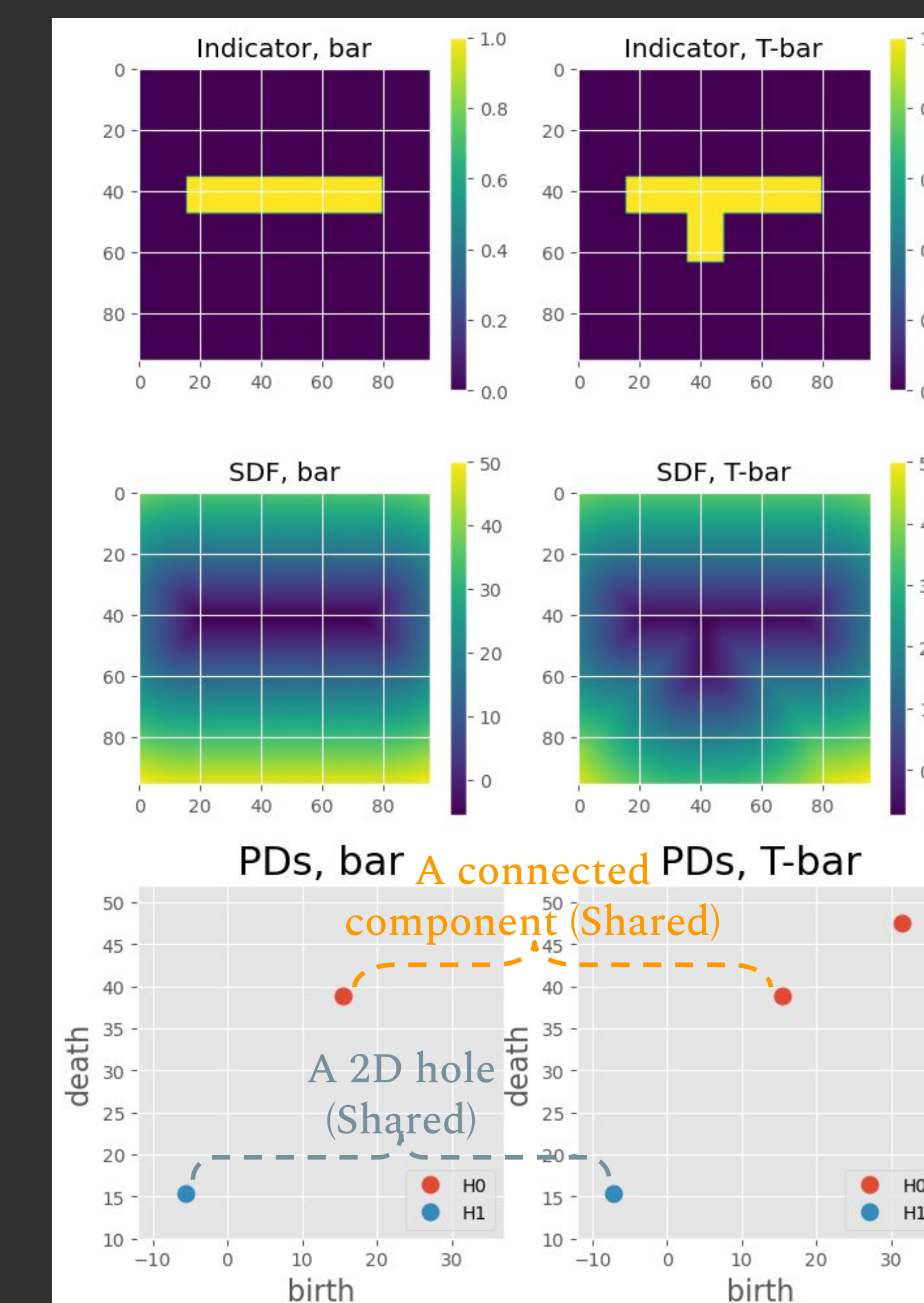
Main Contributions

- A **principled topological regularization** through Persistent Diagrams (PDs), aka Persistent Homology (PH)
- A **generative topology model – PDGen** – capturing topological feature distributions of 3D shapes as **generating parameters on homology groups**
- A **set of novel network layers** to represent and control PDs of SDFs in a differentiable way
- A **study on synergistic generalization** from laplacian smoothing to topological regularization hedging against sparse constraints

Algebraic Preliminaries

- 3D shapes have three homology groups
 - H0: connected components
 - H1: holes in 2D features
 - H2: voids in 3D volumes
- **Direct sum factorization** gives orthogonal basis of vector spaces on topological features
 - Disjoint unions | orthogonality
 - **Coefficients on basis** lead to a **probabilistic model (PDGen)**

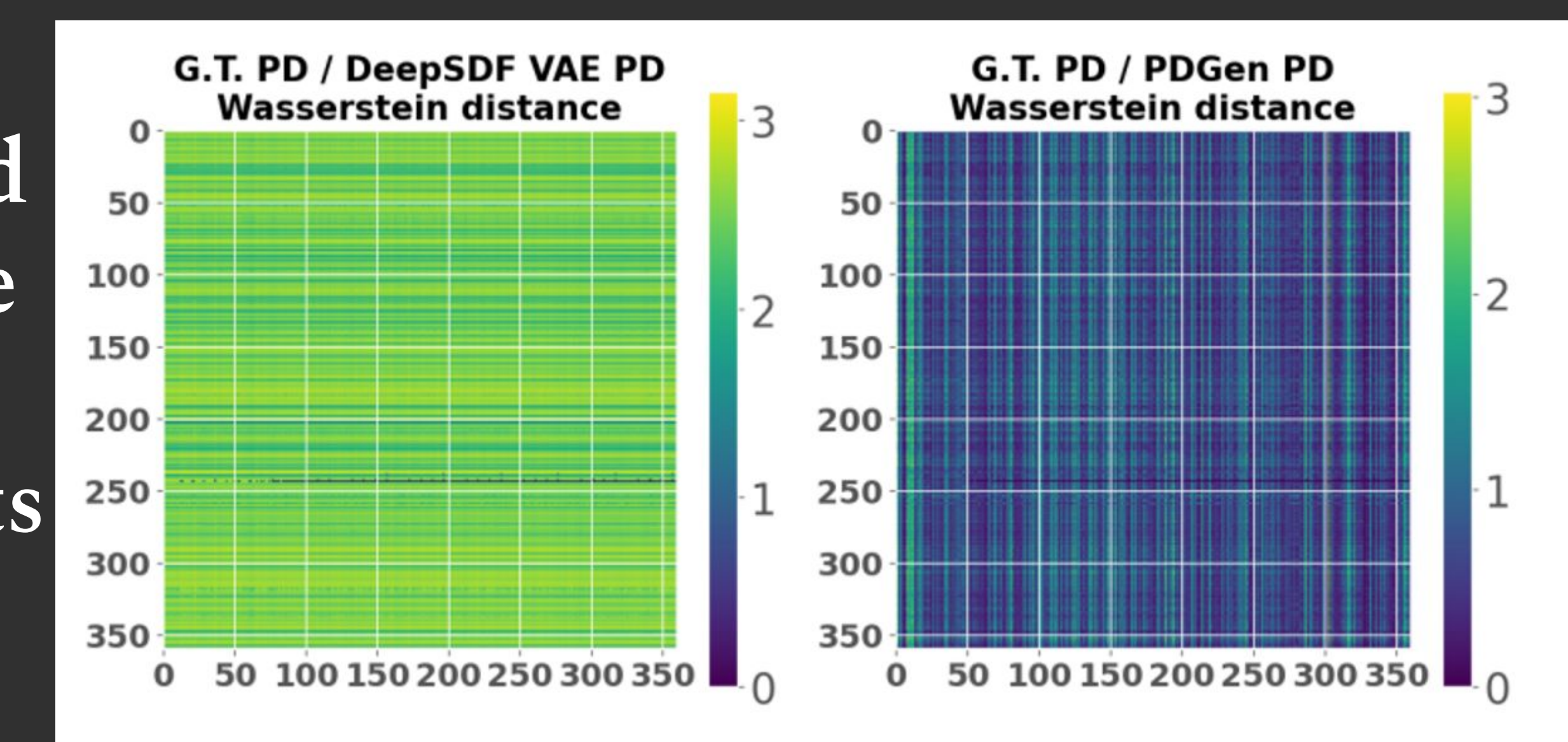
Effective PDs: A 2D Example Pair



Major Results

Drastically improved distribution alignment between Ground-Truth PDs and our PDGen

Color-Coded PD Distance Matrix Between Sets Of Shapes

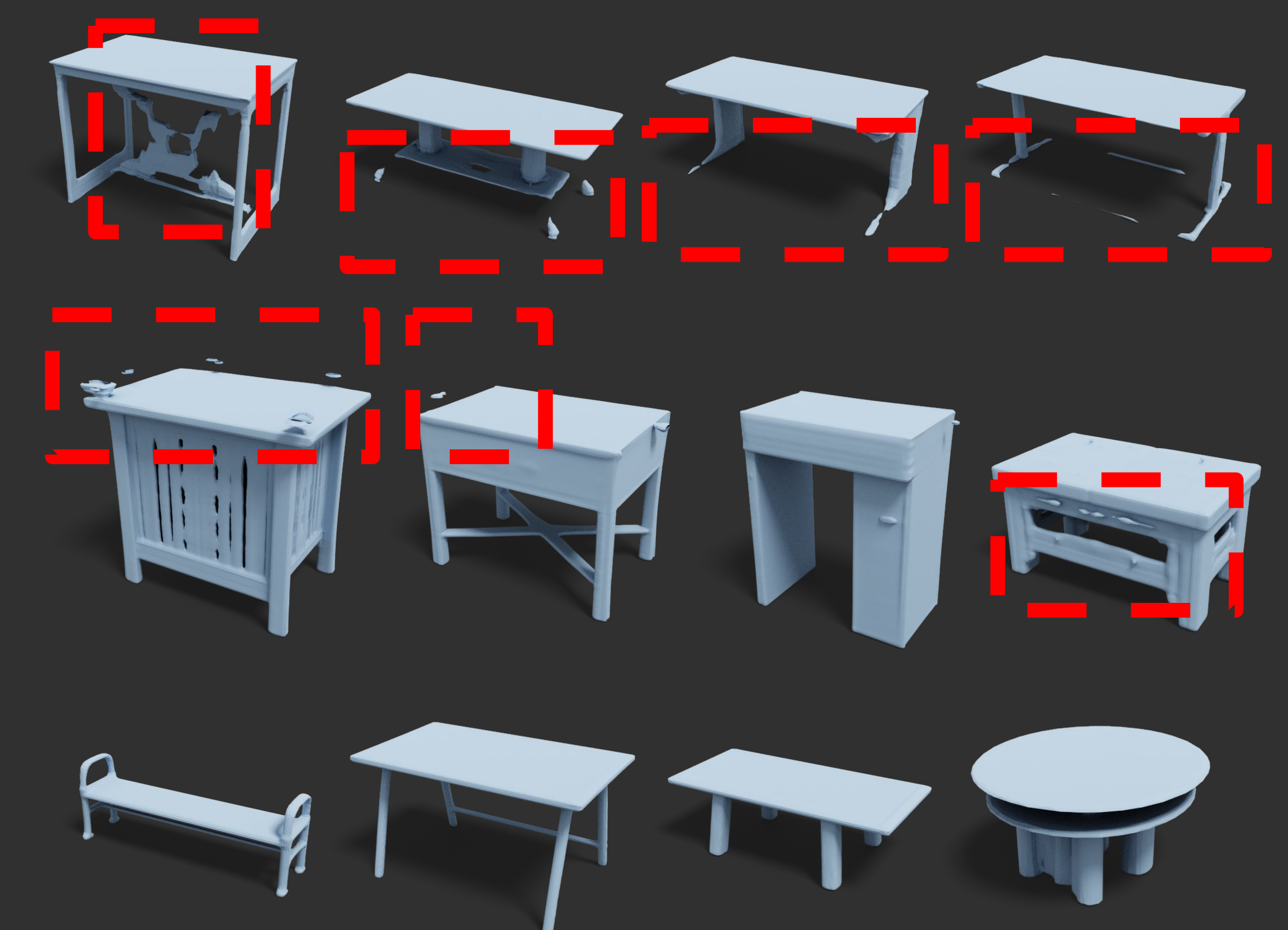


Generation Comparisons

DeepSDF-VAD

3DShape2VecSet

Ours-StEik



Geodesic Interpolation On PDs w/ PDGen

